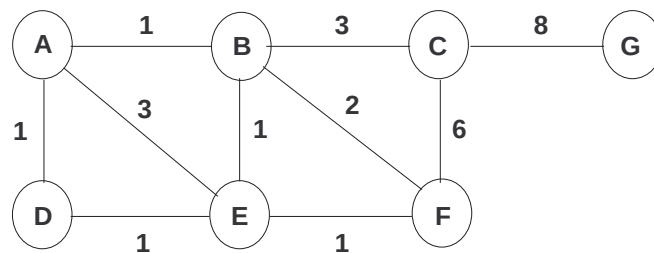


LUNDS TEKNISKA HÖGSKOLA
Institutionen för telekommunikationssystem
Tentamen i DATORKOMMUNIKATION
Datum: 2004-12-16
Sal: MA 10
Tid: 14.00-19.00

- All solutions should be properly written and justified.
 - All calculations have to be shown.
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1.
 - a) To which layers in the OSI model do the following services belong?
 - i) file transfer
 - ii) controlling the dialogue between end system application
 - iii) frame synchronization
 - iv) routing
 - v) frequency modulation
 - b) A twisted pair cable is built up by two copper wires which are twisted around each other in an attempt to avoid electrical/magnetic disturbance. How do such disturbances affect coaxial cables and optical fibers?
 - c) Compute the checksum for the following message: bace19f532. Assume that a checksum of one byte is used. Give your answer in hexadecimal representation.
2.
 - a) Describe the set-up procedure for a packet switched network that is based on
 - i) virtual circuit
 - ii) datagram.
 - b) Mention two differences between ATM and Frame Relay?
 - c) A frame with the length 12 bits should be transmitted from the sender to the receiver. The frame consists of 8 bits data and 4 bits FCS.
 - i) Which P (polynomial) should be used: P=101, P=1011 or P=10011?
 - ii) Which is the bit pattern of the transmitted frame if the message is M=10011001?
3.
 - a) What is the main purpose of *ICMP* packets?
 - b) Give an example on how the CSMA/CD protocol can help in providing QoS (Quality of Service).

- c) What is the probability that at least two frames are correct received over a wireless link if the sender sends 8 frames in total. The probability that one bit is received correct is P_b and each frame is L bits long.
4. a) Explain the following terms
- half duplex
 - peer protocols
 - guided media
- b) ATM uses both virtual channels and virtual paths. What is the relationship between these two? Give an argument to implement both.
- c) Which of the following access methods would you choose to implement in a satellite system? (ALOHA, CSMA, TOKEN)
5. a) What is the difference between a multicast address and a broadcast address?
- b) What are the advantages of dividing an Ethernet LAN with a bridge?
- c) Use Dijkstra's algorithm to compute the shortest path from node F to every other node.



- d) In the IP routing table for a host there are normally two broadcast addresses. Which are they and what are their purposes and limitations?
6. a) Describe each of the following ARQs, **and** also their strength and weaknesses.
- Stop-and wait ARQ.
 - Go-back-N ARQ.
 - Selective repeat ARQ.
- b) What does piggybacking mean?
7. a) In classful IPv4 addressing there are five classes: A, B, C, D and E. Here each address is 32 bits long. Class A addresses starts with

0, class B with 10, class C with 110, class D with 1110 and class E with 11111. For class A, B and C give the length of the Netid and the Hostid in bytes or in bits.

- b) What does classless addressing mean in IP networks?
 - c) For what is the ARP (Address Resolution Protocol) used?
 - d) Name three strengths of the IPv6 compared to the IPv4.
 - e) Name one of the strategies for the coexistence of IPv4 and IPv6 in the Internet.
- 8.
- a) Describe the fundamental differences between UDP (User Datagram Protocol) and TCP (Transmission Control Protocol).
 - b) Describe the functions of the underlined fields in the TCP header.

<u>Source port address</u>					<u>Destination port address</u>				
<u>Sequence number</u>									
<u>Acknowledgement number</u>									
<u>HLEN</u>	Reserved	urg	ack	psh	rst	syn	fin	<u>Window size</u>	
<u>Checksum</u>					<u>Urgent pointer</u>				
Options and padding									

- c) Both transport protocols at OSI level 4 and the datalink protocols at layer 2 do error detection and correction. Why is this mechanism needed on both layers?
- d) Describe the concept of the “Bandwidth-Delay Product”.

Merry Christmas

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Happy New Year !